

MUONIO LAUKUKERO, Finland

WMO: 028200

Lat: 68.063N

Lon: 24.033E

Elev: 2500

StdP: 13.42

Time Zone: 2.00 (EUE)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.3	-2.7	-11.7	3.2	-6.6	-6.8	4.2	-1.7	57.2	12.1	52.8	14.9	27.6	230	1.175

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	68.5	56.3	64.8	54.8	61.4	53.0	58.7	65.5	56.5	62.2	54.5	59.0	13.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
56.4	74.3	59.6	54.0	68.1	57.2	52.2	63.7	55.7	26.6	65.9	25.1	62.2	23.8	59.1	68.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
49.9	44.6	40.2	-12.3	70.4	3.9	3.7	-15.0	73.1	-17.3	75.2	-19.5	77.3	-22.3	80.0
			-9.1	60.9	6.9	3.4	-14.1	63.4	-18.2	65.4	-22.1	67.3	-27.1	69.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.3	12.1	12.5	16.3	24.3	35.5	44.2	51.9	48.3	40.3	27.8	21.1	16.4
	DBStd	15.48	7.61	8.50	6.46	6.52	9.29	8.11	7.62	6.21	5.90	6.15	5.97	6.61
	HDD50	7780	1176	1050	1045	771	463	213	68	106	292	689	866	1041
	HDD65	13027	1641	1470	1510	1221	916	625	410	518	740	1154	1316	1506
	CDD50	233	0	0	0	0	12	38	127	54	2	0	0	0
	CDD65	6	0	0	0	0	0	0	5	1	0	0	0	0
	CDH74	10	0	0	0	0	0	0	9	1	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	20.8	23.7	23.3	24.0	20.1	19.1	19.2	17.1	17.0	19.6	20.7	22.0	23.9
Precipitation	PrecAvg	20.2	1.2	1.0	0.9	1.0	1.2	2.2	3.1	2.9	2.0	1.7	1.5	1.3
	PrecMax	26.9	2.3	2.2	1.8	2.6	3.1	4.1	5.8	5.0	3.7	3.5	3.1	2.4
	PrecMin	15.4	0.4	0.2	0.3	0.4	0.2	0.1	1.1	1.4	0.4	0.6	0.4	0.4
	PrecStd	3.2	0.4	0.5	0.4	0.5	0.8	1.1	1.3	1.1	0.9	0.8	0.7	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.5	32.4	31.5	47.4	67.0	70.1	75.1	67.9	55.3	44.7	39.1	32.0
		MCWB	30.0	31.1	29.4	39.6	50.7	56.0	60.1	56.5	49.6	42.2	36.9	31.8
	2%	DB	27.0	27.7	29.3	40.7	59.9	64.5	70.6	64.3	52.7	40.4	34.3	29.3
		MCWB	26.6	27.3	28.3	35.5	50.1	53.4	57.5	55.4	49.0	39.3	33.8	28.9
	5%	DB	24.8	25.8	27.6	36.0	53.7	60.5	67.0	60.3	50.5	38.7	31.1	26.8
		MCWB	24.5	25.4	26.8	32.9	45.8	50.9	56.2	53.3	47.3	37.7	30.8	26.5
	10%	DB	22.3	23.7	25.5	32.4	48.5	56.7	63.3	56.7	48.2	36.5	28.9	25.0
		MCWB	21.9	23.3	24.9	30.5	42.8	48.7	54.6	51.9	46.2	35.6	28.7	24.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	30.0	31.8	30.3	40.2	52.2	58.6	62.5	60.6	52.9	41.4	36.9	31.9
		MCDB	30.3	32.6	30.7	46.5	61.9	63.2	67.1	64.2	53.4	41.7	37.2	32.1
	2%	WB	26.7	27.2	28.6	36.5	49.8	55.1	59.9	56.7	49.8	39.7	33.5	29.0
		MCDB	27.0	27.5	29.1	40.2	59.2	59.1	66.7	61.0	51.8	40.3	34.8	29.3
	5%	WB	24.4	25.4	26.8	33.0	46.9	52.0	57.9	54.7	48.0	37.9	30.7	26.6
		MCDB	24.6	25.7	27.4	35.4	53.6	56.7	64.9	58.5	49.7	38.7	31.0	26.7
	10%	WB	21.9	23.4	24.9	30.9	42.8	49.3	56.1	52.9	46.4	35.7	28.7	24.8
		MCDB	22.2	23.7	25.5	32.4	48.1	55.3	62.1	56.2	47.6	36.3	28.8	25.0
Mean Daily Temperature Range	5% DB	MDBR	6.3	6.0	6.1	7.1	9.8	11.4	11.1	9.4	7.2	5.6	5.8	5.7
		MCDBR	9.2	7.4	7.3	10.2	14.9	15.7	14.1	12.5	9.7	7.7	8.2	6.9
		MCWBR	9.7	7.4	7.0	7.4	8.7	8.5	7.5	7.3	7.0	6.5	7.6	7.0
	5% WB	MCDBR	9.0	7.1	7.0	9.2	14.4	12.9	12.9	10.5	8.1	6.9	7.9	6.9
		MCWBR	9.4	7.2	6.8	7.1	9.3	8.2	7.9	6.8	6.6	6.2	7.6	7.2
Clear-Sky Solar Irradiance	taub	0.168	0.226	0.262	0.289	0.307	0.312	0.319	0.306	0.279	0.245	N/A	N/A	
	taud	1.917	2.116	2.233	2.262	2.461	2.534	2.518	2.537	2.583	2.343	N/A	N/A	
	Ebn at Noon	41	188	250	274	283	284	277	270	252	185	N/A	0	
	Edh at Noon	3	14	23	31	29	27	27	24	17	11	N/A	0	
All-Sky Solar Radiation	RadAvg	11	156	633	1250	1558	1576	1475	1063	565	187	30	0	
	RadStd	1	17	37	63	152	115	146	80	62	31	3	0	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page